

Concentration Calculations

Types of Concentration		Formula	Common Application
Percent	mass/volume percent		Intravenous solutions, such as a saline drip
	mass/mass percent		Concentration of metals in an alloy
	volume/volume percent		Solutions prepared by mixing liquids
Very Small	Parts per million, ppm		Safety limits for contaminants, such as mercury or lead in food or water
	Parts per billion, ppb		
Molarity or Molar concentration			Solutions used as reactants

Examples

- Find the mass of pure silver in a sterling silver ring that has a mass of 6.45g. Sterling silver has a silver concentration of 92.5% (m/m).
- A sample of lake water has a mass of 310. g and contains 2.24 mg of dissolved oxygen. Calculate the oxygen concentration in parts per million.

- Calculate the molar concentration of 12 g of copper (II) sulfate dissolved in 70 mL of solution
- Find the mass of solute in 28 mL of a 0.045 mol/L solution of calcium hydroxide.
- The concentration of chlorine in swimming pools is generally kept in the range of 1.4-4.0 mg/L. A pool contains 3.0 ppm of chlorine. Is this concentration within the acceptable range? Show your work, and explain your reasoning. (Hints: Assume the density of any dilute aqueous solution is the same as pure water. The density of pure water is 1g/mL)